

CDA 1Tenth: Operating Environment Build Guide

Introduction

Creating a consistent and controllable operating environment is essential for testing and developing the CDA 1Tenth vehicle. The testing space needs to provide a simulation environment that will allow for the CDA1Tenth vehicle functionality and maneuverability to be tested. The final setup consisted of a 12-foot by 12-foot floor made from interlocking foam tiles. The road configuration was assembled using half-inch PVC pipes. Tape was used to secure the layout and provide additional markings. Additional decorations were added to create an immersive environment for the vehicle.

Design Requirements

The operating environment was designed to meet several functional and practical requirements to support testing and development of the CDA1Tenth vehicle. The following criteria were established to ensure the track would be suitable for a wide range of vehicle behaviors and test scenarios:

Two-Lane Traffic Support

The layout needs to accommodate bi-directional vehicle movement, which requires a minimum track width sufficient for two lanes. Each lane needs to allow clearance for a vehicle approximately 7 inches (≈ 178 mm) wide, with additional buffer space to avoid collisions. As a result, the minimum track width was set at 22 inches (≈ 559 mm).

Minimum Turn Radius

To enable realistic and effective maneuverability testing, the track has to support a minimum turn radius of 250 mm (≈ 9.8 inch). This constraint ensured that the vehicle could navigate tighter curves while still being challenged by the layout.

High-Friction, Flat Surface

The driving surface needs to be smooth, level, and high-friction to ensure consistent vehicle traction and minimize skidding during turns and acceleration. EVA foam tiles (2' x 2') were selected due to their ease of modular assembly and texture.

Modularity and Reusability

The environment needs to be fully disassembleable and reconfigurable. All components, including the floor tiles and PVC wall structure, are required to be

easy to transport, store, and reassemble without specialized tools or permanent fixtures.

Support for Multiple Testing Scenarios

The layout is designed to support a variety of vehicle tests, including:

- **Turn Radius Validation:** Curved sections are included to test the vehicle's ability to meet the 250 mm minimum turning radius requirement.
- **Longevity Testing:** A wider circular loop is integrated into the design to allow for continuous driving loops, ideal for long-duration tests assessing wear, thermal behavior, and battery performance.
- **Maneuverability Testing:** Narrow paths, turns, and intersections are included to assess fine steering control, obstacle avoidance, and dynamic path planning.

Bill of Materials

Part #	Item	Description	Quantity	Notes
1	Foam Tiles	2 ft x 2 ft interlocking high-friction floor tiles	36	Covers 12 ft x 12 ft area
2	81 3/8" Pipe	1/2" PVC Pipe	2	For course siding
3	71 1/2" Pipe	1/2" PVC Pipe	1	For course siding
4	52 7/8" Pipe	1/2" PVC Pipe	1	For course siding
5	31.685" Pipe	1/2" PVC Pipe	1	For course siding
6	18" Pipe	1/2" PVC Pipe	6	For course siding
7	11.25" Pipe	1/2" PVC Pipe	1	For course siding
8	5" Pipe	1/2" PVC Pipe	1	For course siding
9	3 1/4" Pipe	1/2" PVC Pipe	1	For course siding
10	56.5" Curved Pipe	1/2" PVC Pipe bent into a 90-degree bend with a 36" radius curve.	2	For course siding.
11	43.7" Curved Pipe	1/2" PVC Pipe bent into a 90-degree bend with a 27.8" radius curve.	5	For course siding
12	18.8" Curved Pipe	1/2" PVC Pipe bent into a 90-degree bend with a 12" radius curve.	4	For course siding
13	5.5" Curved Pipe	1/2" PVC Pipe bent into a 90-degree bend with a 3.8" radius curve.	3	For course siding

14	11" Curved Pipe	½" PVC Pipe bent into a 180-degree bend with a 3.8" radius curve.	9	For course siding
15	Straight Connector	PLA – 3D printed connector with tab	15	For connecting ½" PVC siding
16	Curve to Straight Connector – Right-Handed	PLA – 3D printed connector with tab	8	For connecting a ½" straight piece of siding to a 3.8" radius curved piece of siding
17	Curve to Straight Connector – Left-Handed	PLA – 3D printed connector with tab	8	For connecting a ½" straight piece of siding to a 3.8" radius curved piece of siding
18	Curve Connector	PLA – 3D printed connector with tab	4	For connecting ½" PVC 3.8" radius curved pieces of siding
19	½" Elbow	PVC Connector	2	For connecting siding
20	2" White Gaffer Tape	30 yd roll	1	For securing course siding to the flooring
21	½" White/Yellow Gaffer Tape	60 yd roll	1	For road markings

Assembly

Curving PVC Pipe

There are multiple ways to curve PVC Pipe. One method is the hot sand method. Heat sand in the oven or over a fire until it reaches 400 degrees. Plug the pipe and use a funnel to fill the pipe with the hot sand. Let the pipe warm up for 10-15 seconds and then place it in a mold to cool. A mold can be made by hammering nails into a piece of plywood to create a path with the desired shape of the pipe.

Floor Assembly

On a flat clean surface, lay and interlock the 36 foam tiles to create a 6x6 grid of tiles measuring greater than 12ft by 12ft on each side.

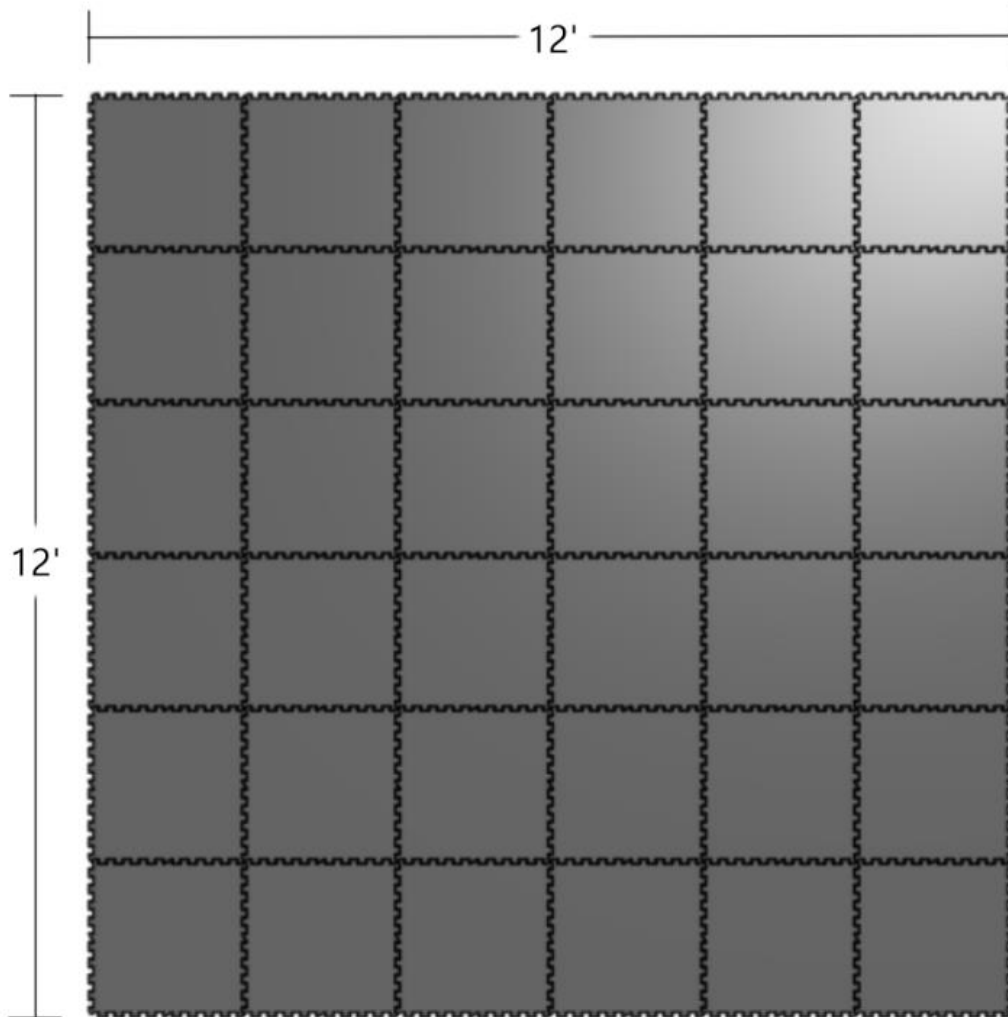


Figure 1: Foam Tile Layout

Course Siding Assembly

Connect the PVC pipes as shown in Figure 2 on top of the foam tile floor. Use the connectors to hold the PVC pipes together. Ensure the tabs are positioned on the outer side of the track, away from the vehicle driving path.

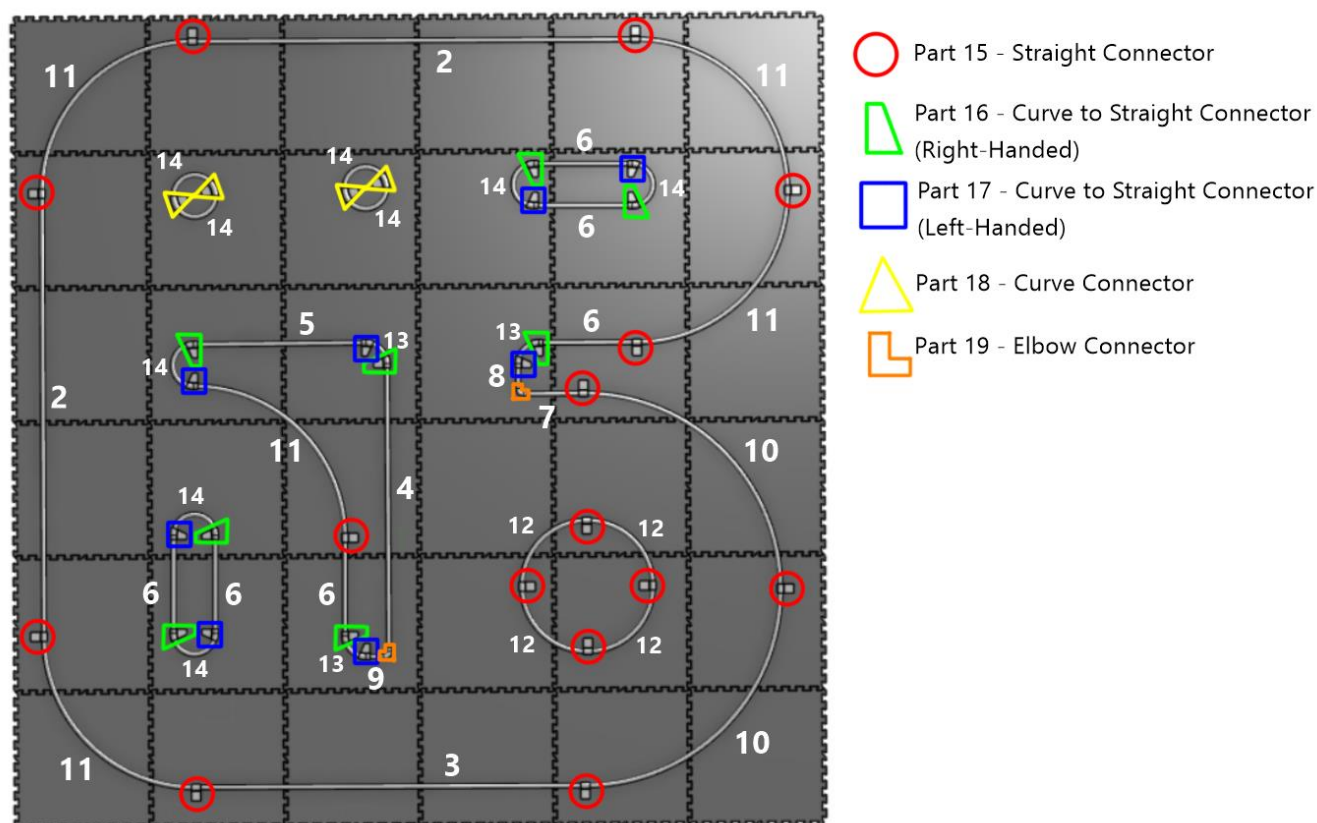


Figure 2: Course Siding Layout

Securing the Course

Tape the PVC Course siding to the foam tile floor by using Gaffer tape overtop of the connector tabs. Use the diagram below to help with placing the islands. There should be a 23" width road across the entire course. From center to center on the PVC pipe should be 24".

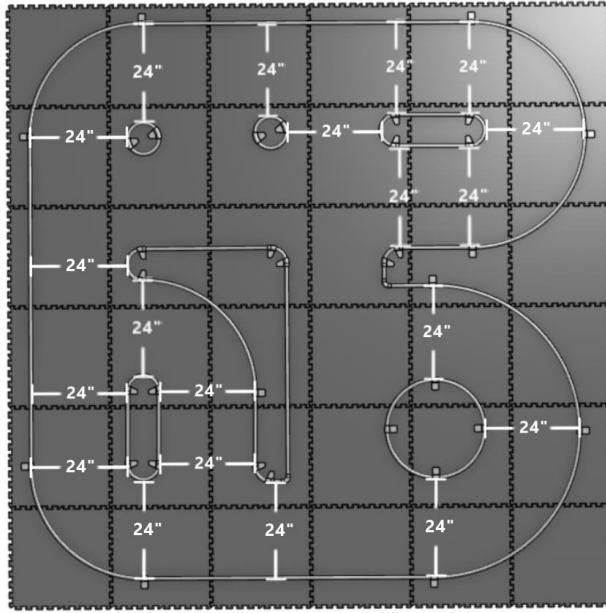


Figure 3: Measuring points on the field

Marking (optional)

Use the ½" yellow or white gaffer tape to mark the lane lines and stop lines as shown in Figure 4.

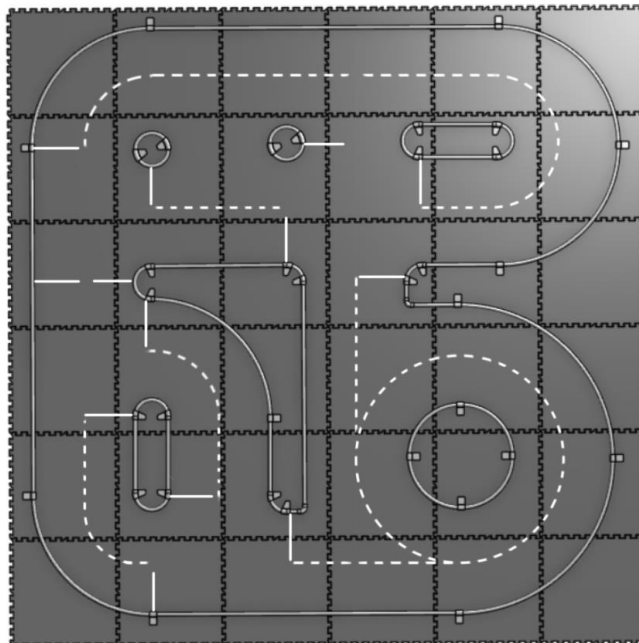


Figure 4: Lane Markings

Additional Features

Stop Signs

To add stop signs to the course, swap out the connector where you want the stop sign with the 'connector with top' version. Then 3D print the sign stand and stop sign using the provided step files. Super glue the stop sign onto the sign stand and press it into the connector on the field.

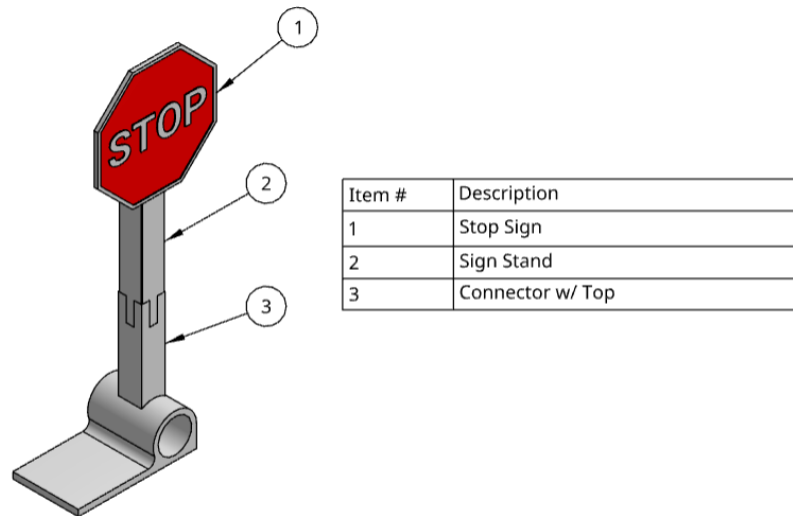


Figure 5: Stop Sign Subassembly

Island Decorations

To enhance the appearance of the course, insert decorated foam pieces into the track islands as shown in the figure below. These foam inserts can be customized with textures, colors, or scenery elements to improve the overall visual appeal.



Figure 6: Track Assembly